

## Slides 4.3: Simple Maintenance and Repairs

Slide outline for Lesson 4.3. Day 1 has thirteen slides; Day 2 has nine.

### Day 1

#### Slide 1: Simple Maintenance and Repairs

- Lesson 4.3, Day 1
- Today: the seven small jobs everyone should know, the tools, and the six steps for fixing anything
- Tomorrow: you do them Image: a tool tray with a screwdriver, pliers, a wrench, a level, a tape measure, and a plunger laid out in a row.

#### Slide 2: Do now

- Which of these have you done or watched: change a bulb, reset a breaker, unclog a drain, stop a running toilet, hang something on a wall, change a filter?
- Write the ones you have done
- What happened the last time something in your home broke?
- Two minutes Notes: Hand vote per job. The count tells you which stations need the most coaching tomorrow.

#### Slide 3: The bill

- Plumber, running toilet, 15 minutes: \$185
- The part: \$8
- Today you learn the fifteen minutes Image: a made-up invoice with a company name blacked out and the \$185 total circled. Notes: The invoice is made up. Say so if asked. The ratio is the point.

#### Slide 4: Job 1, change a light bulb

- Switch off. Let it cool.
- Read the label on the socket: max watts. Never go over.
- Old bulb out counterclockwise. New bulb in clockwise until snug.
- Pick an LED: same light, a tenth of the power Image: a lamp socket with a “60 W MAX” label visible.

#### Slide 5: Job 2, reset a tripped breaker

- Open the panel door. Find the switch sitting in the middle, not on and not off.
- Push it fully OFF. Then push it ON.
- If it trips again right away: stop. Unplug what you just turned on. Tell an adult.

- Dry hands. No tools in the panel. Ever. Image: a breaker panel with a row of switches, one clearly out of line with the others. Notes: Make them say “off, then on.” The middle position will not reset from the middle.

### Slide 6: Job 3, unclog a drain without chemicals

- Pull the hair out first with a plastic drain tool
- Plunge: cover the drain, water over the cup, push and pull six times
- Then hot water
- Never mix chemicals. Never plunge after chemicals; they splash. Image: a bathroom sink with a plastic zip-style drain tool beside it.

### Slide 7: Job 4, stop a running toilet

- Lift the tank lid. Look, do not reach yet.
- Is the chain caught under the flapper? Free it.
- Is the flapper sitting flat over the hole? Reseat it.
- Water too high? Bend the float arm down or turn the float screw. Image: the inside of a toilet tank with the flapper, chain, and float labeled. Notes: This is the example for the six troubleshooting steps on Slide 11.

### Slide 8: Jobs 5 and 6, a loose screw and hanging something

- Screw: match the driver to the head (plus sign is Phillips, slot is flathead). Push in while turning. Stop when snug.
- Wall: find the stud (knock, or a stud finder) or use an anchor
- Mark with a level. Nail or screw at a slight upward angle.
- Not where wires or pipes run: above outlets and beside faucets Image: left, two screw heads side by side; right, a level on top of a picture frame.

### Slide 9: Job 7, clean a filter

- Furnace filter: every one to three months; the arrow points toward the furnace
- Range hood filter: monthly, in the dishwasher or hot soapy water
- Dryer lint trap: every load. Dryer vent: yearly. Lint is a fire.
- Switch the system off first Image: a furnace filter held up to the light, half gray with dust. Notes: Hold up the real filter. Show the arrow.

### Slide 10: Tool ID

- Phillips screwdriver, flathead screwdriver
- Pliers, adjustable wrench
- Level, tape measure
- Sink plunger (cup) and toilet plunger (flange) Image: the seven tools with names under each, and under the names the Turkish, Portuguese, and Spanish words. Notes: Pass the tray, one minute per table. Point to a tool at each table and ask one student to name it.

### Slide 11: Six steps for fixing anything

- a. What is the problem?
  - b. What should it do?
  - c. Test to find the cause
  - d. Fix it
  - e. Check that the fix worked
  - f. Stop it happening again
- Image: the six steps as a loop with the running toilet drawn at each step. Notes: This is the state's troubleshooting process (CTE PSI 3. a)) in plain words. Walk the toilet through all six. Students fill Part 3.

### Slide 12: Safety rules for tomorrow

- Eyes protected at the screw station
  - No tool leaves its station
  - Nothing is plugged in
  - Dry hands at the drain station
  - One person works, the team coaches
- Notes: Students copy all five. These are on the clipboard tomorrow.

### Slide 13: Exit card

- 3 tools you can name
- 2 jobs you could do at home this week
- 1 job you would leave to a professional, and why

## Day 2

### Slide 14: Repair stations

- Lesson 4.3, Day 2
- Five stations, four minutes each, everyone performs once
- The card is the steps. The test is the proof. Image: a wide shot of the five stations set up on tables.

### Slide 15: Do now

- Name the tool: a bubble in a tube
  - Name the tool: a rubber cup on a stick
  - Name the tool: a tip shaped like a plus sign
  - Which station will be hardest? Why?
  - Two minutes
- Notes: One-minute debrief; this is the tool ID check.

### Slide 16: Rotation

- Four minutes, then the reset cue: lights, countdown, hands empty
- Initial your card only after the test

- Safety glasses on at Station 1
- Coach; do not grab Notes: Restate the five rules from yesterday. Start the timer.

### Slide 17: The five stations

- Station 1: drive a screw flush, then remove it
- Station 2: seat the plunger, feel the seal
- Station 3: measure five lines to the nearest eighth
- Station 4: level the bracket; hang the frame straight
- Station 5: find the tripped breaker, say the reset in order; change the bulb Image: five small photos, one per station. Notes: Keep this up during the rotation.

### Slide 18: The maintenance calendar

- Monthly: detectors, furnace filter, range hood, lint trap every load
- Fall: gutters, furnace check, fans reversed, hoses in
- Winter: salt the steps, watch for ice dams
- Spring: roof and gutters, AC test, deck check
- Yearly: batteries, water heater drain, extinguisher gauge, ten-year detectors Notes: Students place every task on the twelve-month grid, Part 4, and justify three.

### Slide 19: Argue it

- Where do the gutters go? Why twice?
- When do the batteries get changed?
- Why do the hoses come in before the first freeze? Notes: Take the two placements teams disagree on most and settle them on the board.

### Slide 20: Exit card

- The six troubleshooting steps in order, from memory
- The station that was hardest, and what made it hard Notes: Sort the cards on the six steps before Lesson 4.5.

### Slide 21: Reset

- Screws back in the box, glasses in the bin, water stays in the bucket
- Tape measures wound in slowly
- Station check before you sit Notes: Same station-check rule as the kitchen. Nobody leaves a station dirty.

### Slide 22: Tomorrow

- Lesson 4.4: where a home's energy and water go, and the math that shows what a fix is worth
- Bring your calculator brain